

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1531

Meyer Rd 138 kV
August 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1531. The Interconnection Customer proposes a 165.0 MW battery energy storage facility interconnecting at the Meyer Rd 138 kV substation in the Virginia area. The request was received on July 25, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 368 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,909,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sorensen to Meyer Rd 138.0 kV	Kanaka to Viking Energy Of Northumberland Substation 138.0 kV	161.5	221	100.0
Bobville to Foothills 138.0 kV	Meyer Rd to Sorensen 138.0 kV L	159.6	204	100.0
Foothills to Youngcane 138.0 kV	Sorensen to Meyer Rd 138.0 kV line	143.6	234	100.0
Kanaka to Youngcane 138.0 kV	Meyer Rd to Sorensen 138.0 kV	140.7	248	100.0
Kanaka to Viking Energy Of Northumberland 138.0 kV L	Sorensen to Meyer Rd 138.0 kV	139.1	257	100.0
Stepdown to Wagley Robertson 138.0 kV	Stepdown to Alpine 138.0 kV	137.4	198	6.8
Alpine to Sorensen 138.0 kV	Viking Energy Of Northumberland to Kanaka 138.0 kV	136.8	237	100.0
Stepdown to Alpine 138.0 kV	Yarnen to McCartney Substation 345.0 kV	128.6	310	16.0
Gila Riv Substation to Cogentrix 138.0 kV	Alpine 345/1kV Transformer	122.1	179	35.1
Harvel to Harrowgate 138.0 kV	North Tower Tap Substation to Cedar Falls Wi 138.0 kV	112.2	192	16.1
Cogentrix to Lone Energy 138.0 kV	Alpine 345/1 kV XFMR	106.6	236	35.1
Bobville to Cochise 138.0 kV	Yarnen to McCartney 345.0 kV	106.0	317	10.5
Poe to Loneenergy 138.0 kV	Alpine 345 kV/1 kV Transformer Bank	105.4	251	35.1
Cedar Falls Wi to Poe 138.0 kV	Alpine 345/1 kV Transformer	104.1	270	35.1
Hardy Reit to Stepdown 138.0 kV	Yarnen to McCartney 345.0 kV	102.8	473	12.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Detaport Substation to Moonstone 138.0 kV	Imperial Savannah Lp to Pheasant Branch 138.0 kV	101.9	298	5.5
Dist 1000 to Birchwood 138.0 kV	Leavitt to Birchwood 345.0 kV	101.6	515	5.7
Trenton Diesel to Dist 1000 138.0 kV	Leavitt to Birchwood 345.0 kV	101.6	527	5.7
Turner Road to Tyler 138.0 kV	Mitchell Dam to Haggin Substation 345.0 kV line	100.1	456	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sorensen to Meyer Rd 138.0 kV	Rebuild line with higher-capacity conductor	\$3,649,000	40% (\$1,479,000)
Bobville to Foothills 138.0 kV	Rebuild line with higher-capacity conductor	\$15,152,000	40% (\$6,140,000)
Youngcane to Foothills 138.0 kV	Rebuild line with higher-capacity conductor	\$16,410,000	40% (\$6,649,000)
Kanaka to Youngcane 138.0 kV	Rebuild line with higher-capacity conductor	\$7,722,000	40% (\$3,129,000)
Kanaka to Viking Energy Of Northumberland 138.0 kV L	Rebuild line with higher-capacity conductor	\$8,989,000	40% (\$3,642,000)
Stepdown to Wagley Robertson 138.0 kV line	Rebuild line with higher-capacity conductor	\$10,535,000	15% (\$1,552,000)
Alpine to Sorensen 138.0 kV	Rebuild line with higher-capacity conductor	\$10,102,000	40% (\$4,093,000)
Stepdown to Alpine 138.0 kV	Rebuild line with higher-capacity conductor	\$7,849,000	12% (\$934,000)
Gila Riv Substation to Cogentrix 138.0 kV	Rebuild line with higher-capacity conductor	\$6,436,000	26% (\$1,668,000)
Harrowgate to Harvell 138.0 kV	Rebuild line with higher-capacity conductor	\$2,868,000	41% (\$1,173,000)
Cogentrix to Lone Energy 138.0 kV	Reconductor line	\$1,470,000	26% (\$381,000)
Cochise Substation to Bobville 138.0 kV	Reconductor line	\$4,046,000	8% (\$321,000)
Poe to Lone Energy 138.0 kV	Reconductor line	\$2,128,000	26% (\$551,000)
Cedar Falls Wi to Poe 138.0 kV	Reconductor line	\$841,000	26% (\$218,000)
Stepdown to Hardy Reit 138.0 kV	Reconductor line	\$4,144,000	9% (\$368,000)
Detaport Substation to Moonstone 138.0 kV	Reconductor line	\$518,000	2% (\$8,000)
Dit 1000 to Birchwood 138.0 kV	Reconductor line	\$6,560,000	3% (\$220,000)
Dist 1000 to Trenton Diesel 138.0 kV	Reconductor line	\$8,351,000	3% (\$281,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Turner Road to Tyler 138.0 kV	Reconductor line	\$3,270,000	3% (\$102,000)
	Total allocated to project		\$32,909,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.